

Strategic Roadmap: Machine Learning Engineering

Codveda Professional Internship X AWS Machine Learning Academy

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Organization: Codveda Technologies
Status: Remote / Self-Paced
Timeline: 20 Apr – 20 May, 2026

EXECUTIVE SUMMARY

This document outlines a high-ROI sprint designed to bridge the gap between academic machine learning theory and industry-standard cloud implementation. By completing the Codveda Level 1 and Level 2 curriculum, the intern will demonstrate proficiency in the full ML lifecycle—data engineering, model optimization, and deployment—while directly satisfying the competency requirements for the **AWS Certified Machine Learning Engineer - Associate** certification.

STRATEGIC ALIGNMENT: CODVEDA ↔ AWS

Domain	Codveda Task	AWS MLA Competency
Data Engineering	Level 1: Preprocessing	SageMaker Data Wrangler / Glue ETL
Supervised Learning	Level 1: Linear Regression	SageMaker Linear Learner / HPO Tuning
Classification	Level 1: KNN Classifier	Instance-based Learning / Model Evaluation
Deep Learning	Level 3: Neural Networks	SageMaker Training Jobs / DL Containers

EXECUTION TIMELINE

Phase I: Foundation & Feature Engineering (Week 1)

- **Task 1.1:** Data Preprocessing (Normalization, One-Hot Encoding, Outlier Treatment).
- **AWS Sync:** Study *Amazon SageMaker Feature Store* and data ingestion patterns.
- **Milestone:** Repository initialization with standardized Python source directory (`/src`).

Phase II: Predictive Modeling & Monitoring (Week 2)

- **Task 1.2:** Linear Regression analysis and residual diagnostics.
- **AWS Sync:** Implement experiment tracking; compare local results vs. *SageMaker Experiments* theory.
- **Milestone:** Technical report on model performance metrics (R^2 , MSE).

Phase III: Classification & Optimization (Week 3)

- **Task 1.3:** KNN implementation with K-Value hyperparameter optimization.
- **AWS Sync:** Analyze model bias/variance trade-offs; map to *SageMaker Clarify* for interpretability.
- **Milestone:** Public technical demonstration (LinkedIn) showcasing the optimization loop.

Phase IV: Final Integration & Stretch Goals (Week 4)

- **Consolidation:** Finalize `README.md`, unit tests, and `requirements.txt` for production readiness.
- **Stretch Goal:** Preliminary implementation of *Neural Networks* (Level 3) using TensorFlow.
- **Milestone:** Formal submission to Codveda and final project presentation.

KEY PERFORMANCE INDICATORS (KPIs)

- ✓ **Technical Debt:** 100% modularized code (no monolithic scripts).
- ✓ **Industry Visibility:** 3 curated LinkedIn updates reflecting AWS/Codveda synergy.
- ✓ **Certification Readiness:** Mastery of 60% of the AWS MLA "Modeling" and "Data Engineering" domains.

- ❗ **Commitment:** I will dedicate 1.5 hours daily to ensure technical excellence and alignment with AWS Academy standards.

Mohamed Yassine Rahal
Machine Learning Intern